

Module 01: Systems — Neural Assemblies, Cortical Organization, and Self-Environment Distinction

Learning Objectives

1. Describe how the brain implements the system/environment distinction through neural organization, from cortical columns to large-scale networks.
2. Explain how the Markov Blanket framework maps onto the brain-body-environment loop, with sensory cortex and motor cortex as the blanket states.
3. Analyze empirical evidence (fMRI resting-state networks, lesion studies) for the brain as a self-organizing system.

Introduction

The brain is the most complex system we know — approximately 86 billion neurons organized into hierarchical assemblies, from individual synapses to cortical columns to large-scale networks. How does this biological system implement the Active Inference framework introduced philosophically in Course 1? This module bridges the gap between abstract theory and neural reality.

Key Concepts

1. Cortical Columns as Functional Units

The neocortex is organized into **cortical columns** — vertical assemblies of approximately 100 neurons that share functional properties. Vernon Mountcastle (1957) discovered that neurons within a column respond to the same sensory features. In Active Inference terms, each column can be understood as a local inference engine, maintaining predictions about a specific feature of the environment and generating prediction errors when those predictions are violated.

The hierarchical organization of the cortex — from primary sensory areas (V1, A1, S1) to higher association areas (prefrontal cortex, temporal pole) — corresponds to the hierarchical

depth of the generative model. Lower levels predict raw sensory features; higher levels predict abstract, temporally extended patterns.

2. The Brain-Body-Environment Loop

The Markov Blanket of the brain is not the skull but the sensory and motor surfaces of the body. **Sensory neurons** (retina, cochlea, mechanoreceptors) constitute the sensory blanket states — they mediate the influence of the external world on internal (neural) states. **Motor neurons** (alpha motor neurons, autonomic effectors) constitute the active blanket states — they mediate the brain's influence on the external world.

This embodied loop — world → sensory → brain → motor → world — is the physical implementation of the perception-action cycle central to Active Inference. Friston (2005) showed that resting-state fMRI activity in the default mode network reflects the brain's ongoing generative model — the "priors" that the brain maintains even in the absence of external stimulation.

3. Self-Organization and Resting-State Networks

Resting-state fMRI studies have identified several large-scale networks that spontaneously organize even when the subject is not performing any task. The **default mode network (DMN)**, **salience network**, and **central executive network** can be understood as components of the brain's generative model operating at different hierarchical levels and timescales.

The DMN, active during rest and self-referential thought, may encode the agent's model of itself — the "self" as a prediction about one's own persistence. The salience network, anchored by the anterior insula and dorsal anterior cingulate cortex, may implement **precision weighting** — determining which prediction errors are amplified and which are suppressed.

Clinical Connections

- **Anosognosia** (unawareness of deficit after stroke) may result from a failure to update the generative model's self-component — the brain continues to predict normal function despite neurological damage.

- **Depersonalization** may reflect low precision on self-model predictions — the agent's "belief" in its own existence becomes uncertain.

Conclusion

The brain implements the Active Inference framework through hierarchical neural assemblies, with the sensory and motor surfaces of the body forming the Markov Blanket that mediates all interchange with the environment. Understanding this neural architecture is essential for connecting theory to empirical evidence, clinical conditions, and experimental paradigms explored throughout this course.